Lesson plan · for the teacher

Data all around the school

Data engineering · Level 1 · Lesson 1

Time
40 min

Materials
Board and chalk; A number card for each pupil (1, 2, 3…); Optional: a phone or computer with a spreadsheet app

Learning goals

- Say what data is, and find it in our school.
- Read a table: rows, columns and cells.
- Build a tidy table of our class, and count from it.

Activity: Our class in a table

1. Warm-up (5 min): Ask pupils what the school writes down every day: the register, marks, the rain gauge, library cards. List their ideas on the board and circle the ones kept as tables.
2. Teach (8 min): Draw a small register on the board. Point to a row (one pupil), a column (one day) and a cell (one letter, P or A). Then show a messy version, with a Total row and a cell holding two values, and ask what goes wrong when we count.
3. Build (15 min): Draw a table with the columns “Pupil no.”, “How I come to school” and “Minutes”. Agree the codes first (W walk, B bus, C bicycle, M motorbike). Each pupil comes up and fills in their row, using their number, not their name.
4. Count (7 min): Count each code with tally marks beside the table, then count the rows with more than 30 minutes. Check that the counts add up to the number of pupils present.
5. Wrap-up (5 min): Ask which rules made the table easy to count: one row per pupil, one fact per column, one value per cell. Then give the exit ticket.

Give each pupil a number before the lesson, and keep the list of numbers and names to yourself. If some pupils don’t know their minutes, accept “about” values and say so on the table.

Discussion

- Which data does our school keep that nobody ever counts?
- Why did we use numbers instead of names in our table?
- What went wrong when the table had a Total row in it?

Exit ticket

- Draw a tiny table with two rows and two columns about anything you like, and label a row, a column and a cell.

Try it in class tomorrow
Draw a table on the board: one row for each pupil, by number, not by name, and columns for how they come to school and the minutes it takes. Then count together: how many walk more than thirty minutes?

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **A table of how pupils come to school and how long it takes: what should one row be?**

◯ One way of travelling ◯ The whole class ◯ One pupil

2 **Match each value from a class table to its kind.**

25 (minutes to walk to school)	• • • •	A date: written one agreed way
Walks / bus / bicycle	• • • •	Text: read by people
2 March 2026	• • • •	A number: it can be added and averaged
“Always helps others”	• • • •	A category: it can be counted

3 **A Total row inside the data helps a computer add up the class.**

◯ True ◯ False

4 **Which of these break the rules of a tidy table? Choose all that apply.**

☐ “14 / 17” in one cell ☐ The class written once, with the cells below left blank

☐ A column headed “Minutes to school” ☐ A Total row between the classes

☐ One row for each pupil

Answers · for the teacher

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1 **One pupil**

The table is about pupils, so each row is one pupil.

2 **25 (minutes to walk to school) — A number: it can be added and averaged**
Walks / bus / bicycle — A category: it can be counted
2 March 2026 — A date: written one agreed way
“Always helps others” — Text: read by people

Numbers like minutes can be added and averaged; categories like a way of travelling are counted; dates need one agreed format; text such as a comment is read, not counted.

3 **False**

A computer treats the Total row as one more pupil with a huge mark, so counts and averages come out wrong.

4 **The class written once, with the cells below left blank; A Total row between the classes; “14 / 17” in one cell**

Blank class cells, Total rows and two values in one cell all confuse a computer.

For you to think about

Which data does your school already keep that could answer a real question, if it were in one tidy table?